Lecture 5-2: OLS Assumptions

Goal: Examine OLS assumptions, omitted variable bias, and multicollinearity.

- Multiple OLS assumptions
  - Linearity in parameters
  - Random sampling
  - No perfect collinearity
  - Zero conditional mean
  - Homoskedasticity
  - Normality
- Omitted variable bias
  - Excluding a relevant variable
  - Bias direction depends on correlations
  - Confounders defined
- Multicollinearity
  - Variance of OLS estimators & estimation
  - Detecting multicollinearity
  - Variance inflation factor
  - Including an irrelevant variable: no bias, but larger variance

Worksheet

1. (*Walking through the unbiasedness proof*) Let the true model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$ satisfy Assumptions I–IV. Let $\hat{r}_{1,i}$ be the residuals from the OLS regression of x_1 on x_2 and a constant. The OLS slope on x_1 in the regression of y on $(1, x_1, x_2)$ can be written as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} \hat{r}_{1,i}\, y_i}{\sum_{i=1}^{n} \hat{r}_{1,i}^2}.$$

   (a) Show that $\sum_i \hat{r}_{1,i} = 0$, $\sum_i \hat{r}_{1,i} x_{2,i} = 0$, and $\sum_i \hat{r}_{1,i} x_{1,i} = \sum_i \hat{r}_{1,i}^2$.

   (b) Plug in $y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + u_i$ and show that $\hat{\beta}_1 = \beta_1 + \dfrac{\sum_i \hat{r}_{1,i} u_i}{\sum_i \hat{r}_{1,i}^2}$.

   (c) Take expectations given X to show that $E[\hat{\beta}_1 \mid X] = \beta_1$. Which assumption does each step use?

2. (*Applied OVB sign and magnitude*) Suppose the true DGP is $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u$ with $\beta_2 = 0.8$. In your sample, $SST_1 = n \cdot MSD(X_1)$ and the auxiliary regression of X_2 on $(1, X_1)$ yields slope $\tilde{\delta} = 0.5$ and $R^2 = 0.64$. You omit X_2 and estimate Y on $(1, X_1)$ only.

   (a) What is the sign and magnitude of the OVB on $\tilde{\beta}_1$?

   (b) If the true $\beta_1 = 1.0$, what is $E[\tilde{\beta}_1 \mid X]$?

   (c) If instead $\tilde{\delta} = -0.3$ with the same β_2, what happens?

3. **(Optional)** (*Two–regressor OVB algebra*) Let the true model satisfy Assumptions I–IV: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. Suppose we estimate the underspecified model $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$. Regress x_2 on x_1 with intercept to get $x_2 = \tilde{\gamma}_0 + \tilde{\delta} x_1 + v$, where v are residuals orthogonal to 1 and x_1. Show that

$$E[\tilde{\beta}_1 \mid X] = \beta_1 + \beta_2 \tilde{\delta},$$

identify the two conditions under which $E[\tilde{\beta}_1 \mid X] = \beta_1$, and interpret $\tilde{\delta}$.